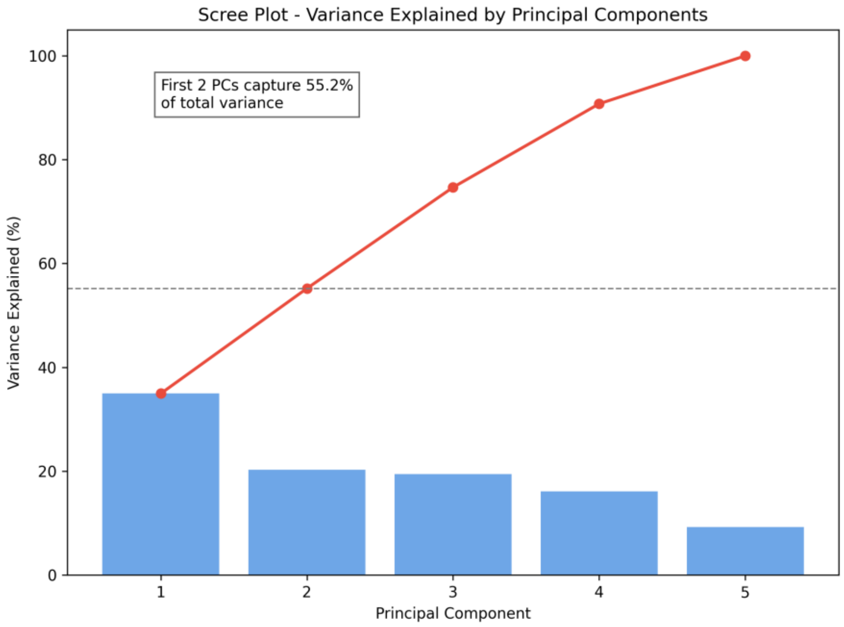
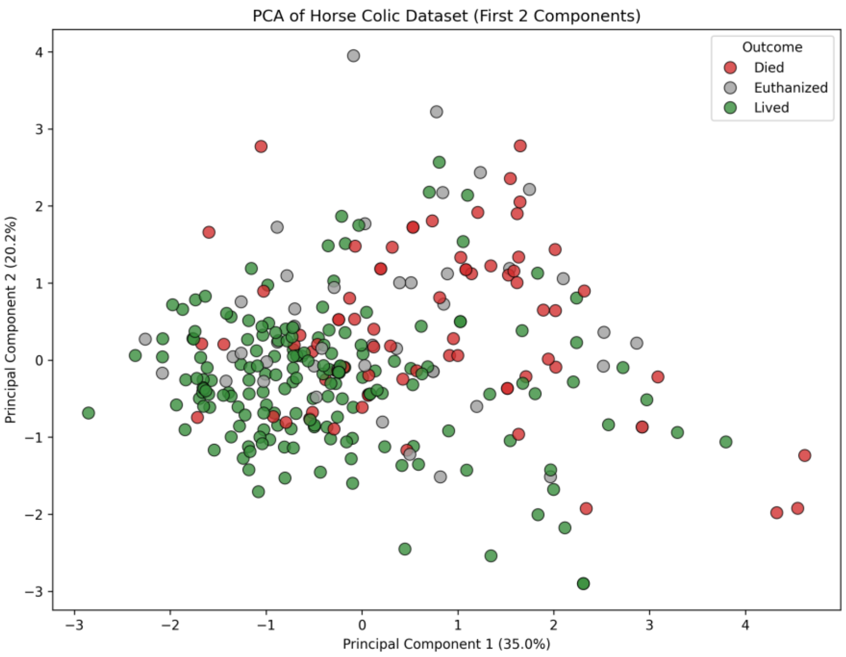
HOMEWORK 6: DIMENSIONALITY REDUCTION

Tomáš Války

1. PCA RESULTS

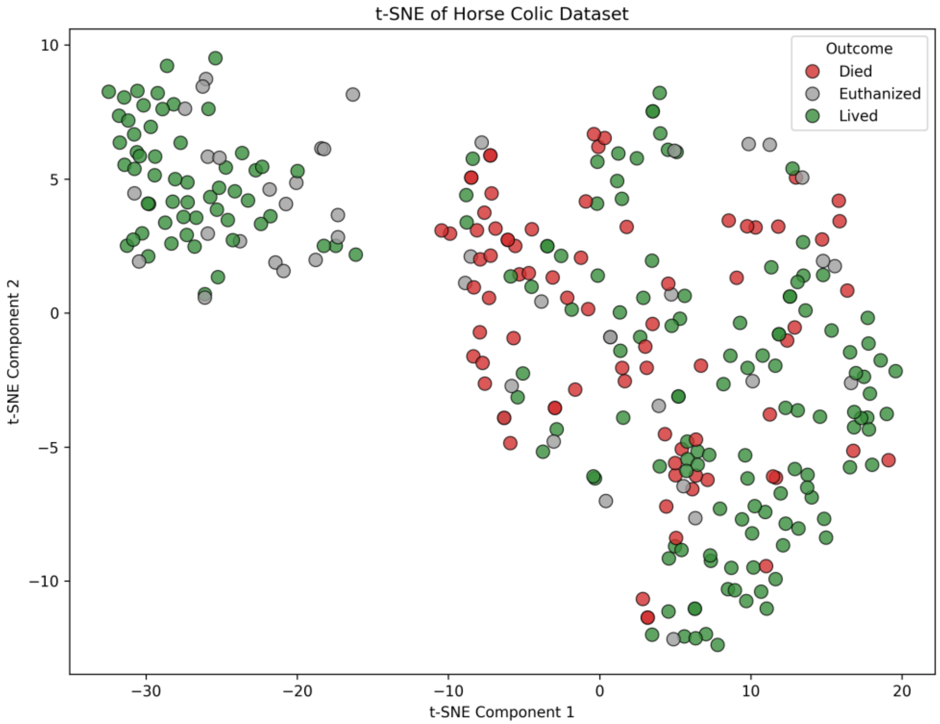
- Data preparation: Extracted 5 numeric variables (pulse, rectal_temp, respiratory_rate, total_protein, packed_cell_volume).
- Imputed missing values using the median strategy and normalized using StandardScaler.
- PCA reduces dimensionality linearly. The Scree plot shows how much total variance is captured by the components.
- A 2D scatter plot of the first two principal components shows minimal clear clustering based on the 'Outcome' variable.





2. ALTERNATIVE METHOD: T-SNE

- t-Distributed Stochastic Neighbor Embedding (t-SNE) was used as the alternative.
- Parameters: n_components = 2, perplexity = 30.
- Unlike PCA, t-SNE is a non-linear technique focusing on keeping similar instances close together in lower dimensions.
- The resulting clusters represent local data groupings, but outcomes remain heavily interspersed.



3. COMPARISON AND DISCUSSION

- PCA vs t-SNE Comparison:
- - PCA is faster and preserves global data structure and variance. However, the first two components did not achieve strong group separability by vital signs.
- - t-SNE captured local non-linear structures but still struggles to cleanly separate Lived, Died, and Euthanized outcomes, implying outcomes are heavily dependent on mixed unmeasured or categorical variables.

Discussion:

- Dimensionality reduction on just 5 blood/vital indicators highlights that horses suffering from colic share highly overlapping physiological distress markers regardless of their final outcome.

5. TOOLS, LIBRARIES & AI SUPPORT

- Programming Language: Python 3.12
- Tools & Libraries:
- - pandas & numpy for data loading and preprocessing
- - scikit-learn for imputation, standardization, PCA, and t-SNE
- - matplotlib & seaborn for generating 2D scatter and bar plots
- - python-pptx for automated presentation generation
- AI Support:
- - Used Google Deepmind Gemini 3.1 Pro (High) via Antigravity Agent to plan and automate both the data analysis pipeline and the complete generation of this presentation deck.